Setting up the ESP IDF 5.5.2 Development Environment on Windows and Compiling Xiaozhi

If any boss needs various product development, ESP32 custom development, or Xiaozhi secondary development, please add WeChat: Kevincoooool. Free guidance services are not provided due to limited energy.

If you need an ESP32 development board, you can visit https://item.taobao.com/item.htm?id=681702043224

1. Download the offline installation package

First, enter

https://dl.espressif.com/dl/esp-idf/

If you are unable to download, you can also access the Baidu Cloud link:

https://pan.baidu.com/s/11jwTFJbcOsV2K_IlkBriJQ?pwd=qm34 Extraction code: qm34

Versions 5.3.1, 5.3.2, and 5.3.3 were available in versions prior to 1.6.5, but the latest code requires 5.4.1 because it adds support for cameras!!!

The latest Xiaozhi requires IDF5.5.2!

This link is the official download website of Espressif, where you can download the offline installation package. Click to download version 5.5.2.

Various versions can also be selected at the bottom of the webpage

Do not compile using VSCODE's plugins!!! It is very error-prone!!! Compile using the following command line!!!

ESP-IDF Windows Installer Download

Open Source IoT Development Framework for ESP32

Universal Online Installer 2.3.5
Windows 10, 11
Size: 4.37 MB

Espressif-IDE 3.1.0 with ESP-IDF v5.3.1 - Offline Installer
Windows 10, 11
Size: 1.91 GB

ESP-IDF v5.5 - Offline Installer
Windows 10, 11
Size: 1.51 GB

ESP-IDF v5.4.2 - Offline Installer
Windows 10, 11
Size: 1.51 GB

ESP-IDF v5.3.3 - Offline Installer
Windows 10, 11
Size: 1.42 GB

ESP-IDF v5.2.5 - Offline Installer
Windows 10, 11
Size: 1.41 GB

Windows 10, 11
Size: 1.06 GB

ESP-IDF v5.0.7 - Offline Installer
Windows 10, 11
Size: 0.96 GB

Installation instructions: ESP-IDF documentation and Espressif Systems Youtube channel.

Want new installer features sooner?

Get Beta version.

Download links to available releases and mirrors.

Release version	Release date		Release notes
Online Installer v2.3	2024-10-09	Download / Mirror - 4.35 MB	Release Notes
ESP-IDE v3.1.0-with-esp-idf-5.3.1	2024-10-07	Download / Mirror - 1.52 GB	Release Notes
Offline Installer v5.2.3	2024-09-27	Download / Mirror - 1.02 GB	Release Notes
Offline Installer v5.3.1	2024-09-09	Download / Mirror - 1.02 GB	Release Notes
Offline Installer v5.2.2	2024-08-29	Download / Mirror - 1.01 GB	Release Notes
Online Installer v2.28	2024-08-29	Download / Mirror - 4.37 MB	Release Notes
Offline Installer v5.3	2024-08-29	Download / Mirror - 1.02 GB	Release Notes
Offline Installer v5.2.2	2024-08-29	Download / Mirror - 1 GB	Release Notes

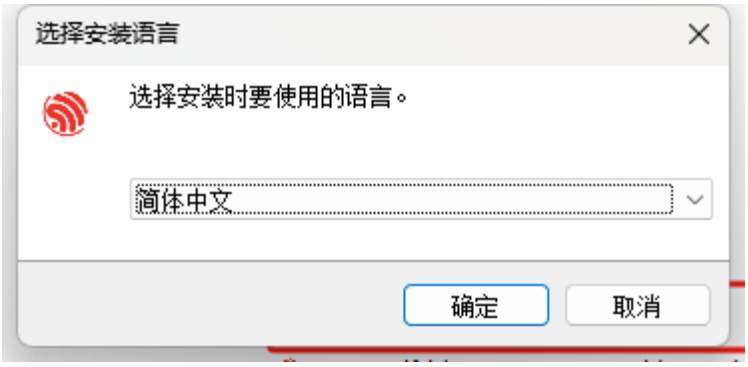
ESP-IDF Windows Installer Support ESP-IDF GitHub

Select any version you need to install

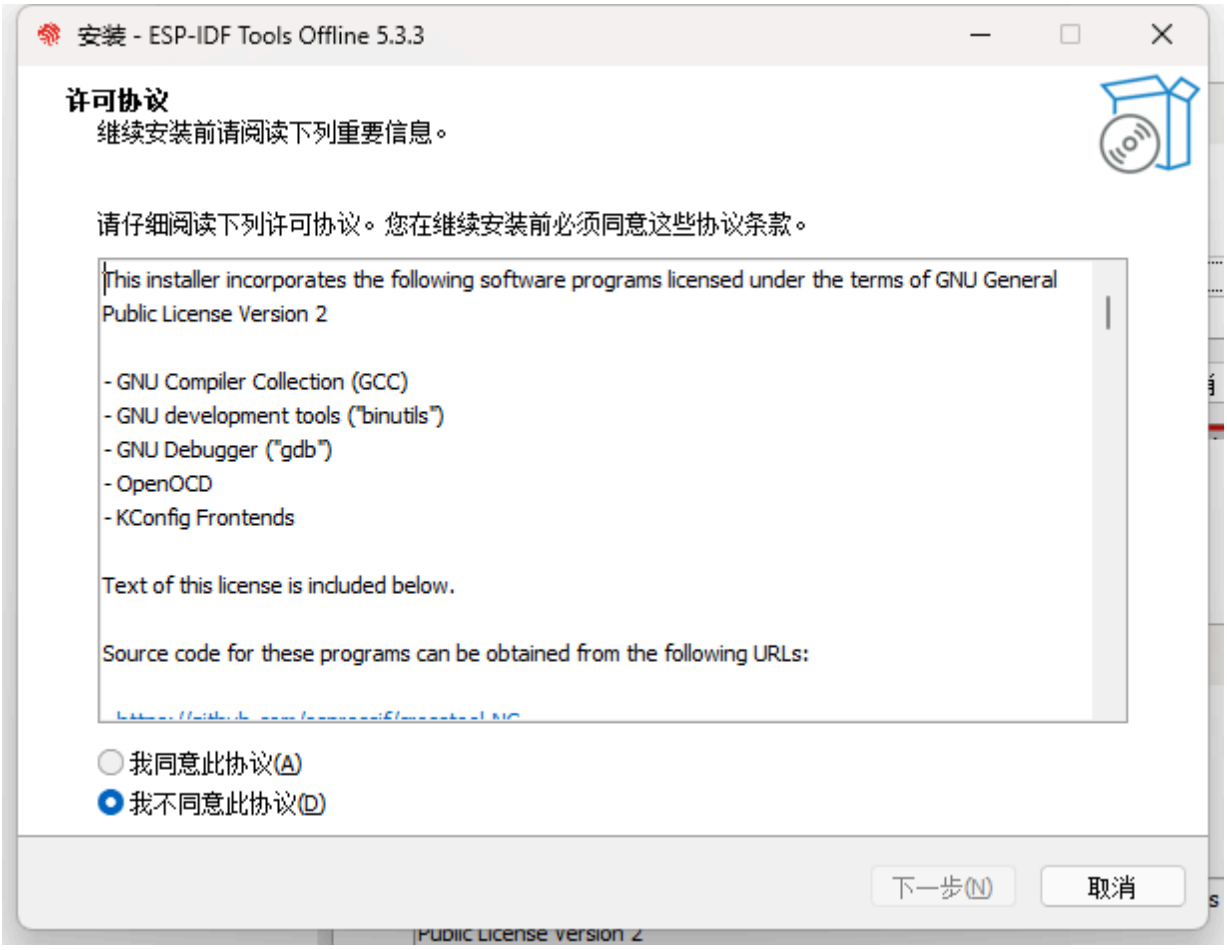
2. Installation Environment

Select the version you need to install, double-click the EXE file to start the installation

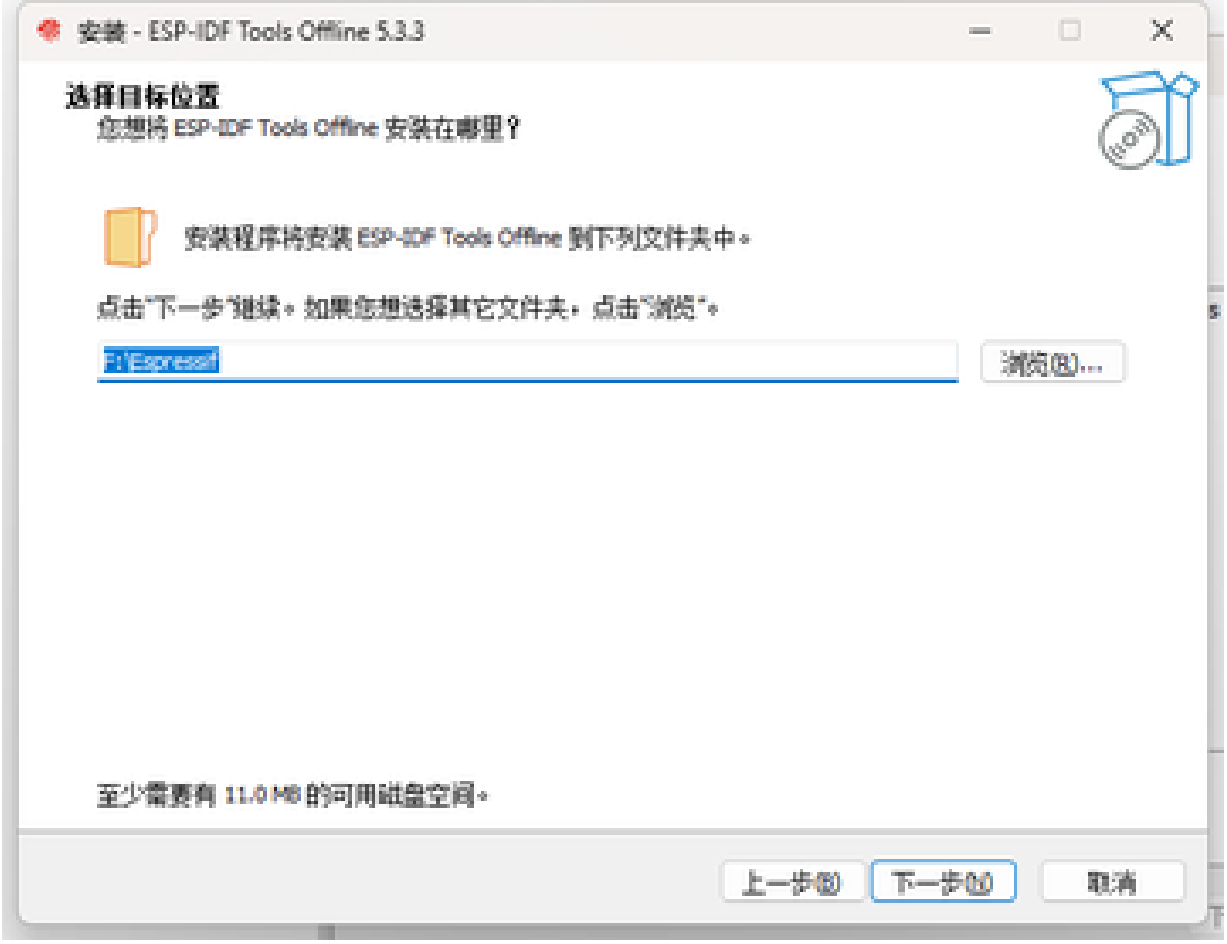
esp-idf-tools-setup-offline-5.5.exe



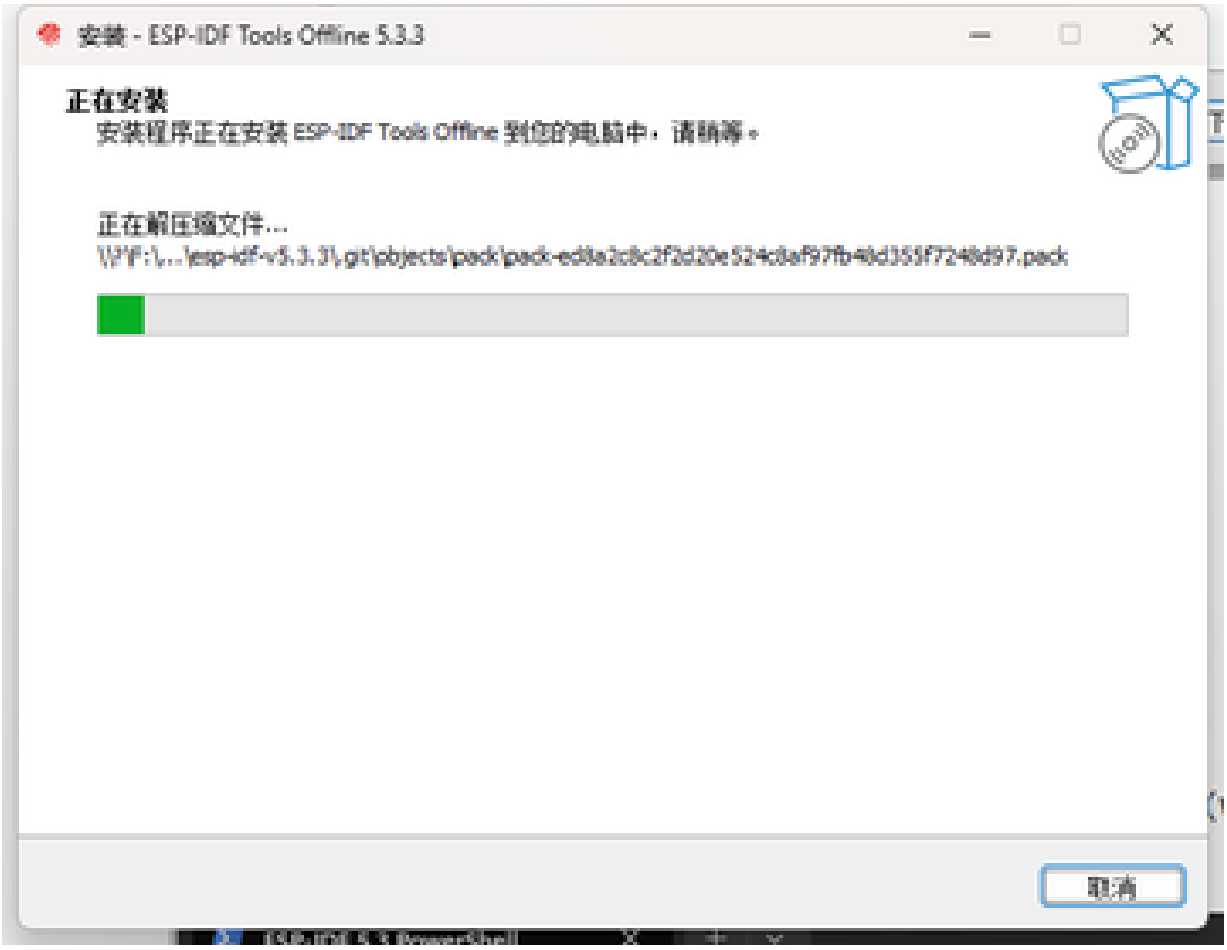
After checking "I agree", just keep clicking "Next".



You can set the folder where your IDF will be installed here. Try not to install it on the C drive.

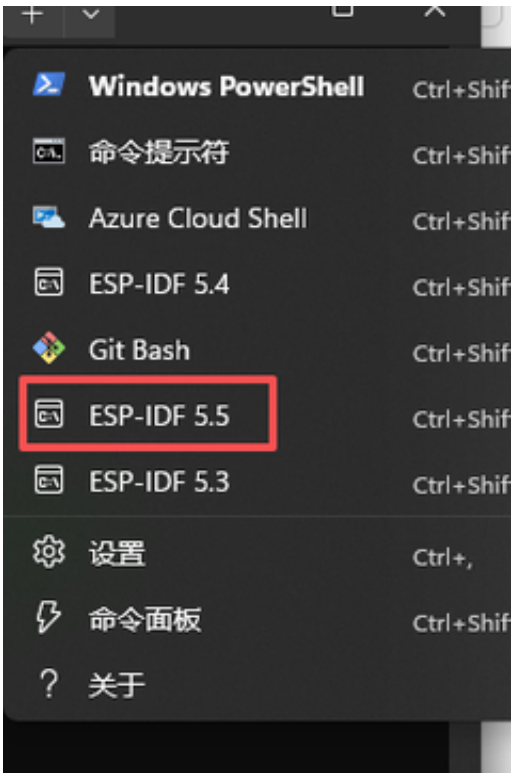


Keep clicking "Next" until the installation is complete, and wait for the installation to finish



3. Test whether the installation was successful

Double-clicking Powershell on the desktop will automatically import the idf environment. For computers with Windows Terminal installed (it comes pre-installed on Win11, and Win10 users can search for the Store to open and install it), they can directly open the terminal, and the + sign in the terminal can directly run the idf environment.



```
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安装最新的 PowerShell，了解新功能和改进！https://aka.ms/PSWindows

Setting PYTHONNOUSERSITE, was not set
Using Python in F:\Espressif\python_env\idf5.5_py3.11_env\Scripts
Python 3.11.2
Using Git in F:\Espressif\tools\idf-git\2.44.0\cmd
git version 2.44.0.windows.1
Activating ESP-IDF 5.5
Setting IDF_PATH to 'F:\Espressif\frameworks\esp-idf-v5.5'.
* Checking python version ... 3.11.2
* Checking python dependencies ... OK
* Deactivating the current ESP-IDF environment (if any) ... OK
* Establishing a new ESP-IDF environment ... OK
* Identifying shell ... powershell.exe
* Detecting outdated tools in system ... Found tools that are not used by active ESP-IDF version.
For removing old versions of ccache, cmake, dfu-util, esp-clang, esp-rom-elfs, esp32ulp-elf, idf-driver, idf-python-wheel
ls, ninja, openocd-esp32, riscv32-esp-elf, riscv32-esp-elf-gdb, xtensa-esp-elf, xtensa-esp-elf-gdb, xtensa-esp32-elf,
xtensa-esp32s2-elf, xtensa-esp32s3-elf use command 'python.exe F:\Espressif\frameworks\esp-idf-v5.5\tools\idf_tools.py u
ninstall'
To free up even more space, remove installation packages of those tools.
Use option python.exe F:\Espressif\frameworks\esp-idf-v5.5\tools\idf_tools.py uninstall --remove-archives.

Done! You can now compile ESP-IDF projects.
Go to the project directory and run:

  idf.py build

PS F:\Espressif\frameworks\esp-idf-v5.5>
```

When the text "idf.py build" appears, it means the installation is successful and the firmware can be compiled.

Run `cd.\examples\get-started\hello_world\` to enter the project directory

```
* Deactivating the current ESP-IDF environment (if any) ... OK
* Establishing a new ESP-IDF environment ... OK
* Identifying shell ... powershell.exe
* Detecting outdated tools in system ... Found tools that are not used by active ESP-IDF version.
For removing old versions of ccache, cmake, dfu-util, esp-clang, esp-rom-elfs, esp32ulp-elf, idf-driver, id
ls, ninja, openocd-esp32, riscv32-esp-elf, riscv32-esp-elf-gdb, xtensa-esp-elf, xtensa-esp-elf-gdb, xtensa-
xtensa-esp32s2-elf, xtensa-esp32s3-elf use command 'python.exe F:\Espressif\frameworks\esp-idf-v5.5\tools\i
ninstall'
To free up even more space, remove installation packages of those tools.
Use option python.exe F:\Espressif\frameworks\esp-idf-v5.5\tools\idf_tools.py uninstall --remove-archives.

Done! You can now compile ESP-IDF projects.
Go to the project directory and run:

  idf.py build

PS F:\Espressif\frameworks\esp-idf-v5.5> cd .\examples\get-started\hello_world\
```

Input `idf.py build`

```
Go to the project directory and run:

  idf.py build

PS F:\Espressif\frameworks\esp-idf-v5.5> cd .\examples\get-started\hello_world\
PS F:\Espressif\frameworks\esp-idf-v5.5\examples\get-started\hello_world> idf.py build
```

Compilation will now begin. Remember to turn off all antivirus software during compilation, including 360, Huorong, Windows Defender, etc., and uninstall PC Manager! This will significantly improve compilation speed.

```
-- Component paths: F:/Espressif/frameworks/esp-idf-v5.5/components/app_update F:/Espressif/frameworks/esp-idf-v5.5/comp
onents/bootloader F:/Espressif/frameworks/esp-idf-v5.5/components/bootloader_support F:/Espressif/frameworks/esp-idf-v5.
5/components/cxx F:/Espressif/frameworks/esp-idf-v5.5/components/efuse F:/Espressif/frameworks/esp-idf-v5.5/components/e
sp_app_format F:/Espressif/frameworks/esp-idf-v5.5/components/esp_bootloader_format F:/Espressif/frameworks/esp-idf-v5.5
/components/esp_common F:/Espressif/frameworks/esp-idf-v5.5/components/esp_driver_gpio F:/Espressif/frameworks/esp-idf-v
5.5/components/esp_hw_support F:/Espressif/frameworks/esp-idf-v5.5/components/esp_mm F:/Espressif/frameworks/esp-idf-v5.
5/components/esp_partition F:/Espressif/frameworks/esp-idf-v5.5/components/esp_pm F:/Espressif/frameworks/esp-idf-v5.5/c
omponents/esp_rom F:/Espressif/frameworks/esp-idf-v5.5/components/esp_security F:/Espressif/frameworks/esp-idf-v5.5/comp
onents/esp_system F:/Espressif/frameworks/esp-idf-v5.5/components/esp_timer F:/Espressif/frameworks/esp-idf-v5.5/compone
nts/esptool_py F:/Espressif/frameworks/esp-idf-v5.5/components/freertos F:/Espressif/frameworks/esp-idf-v5.5/components/
hal F:/Espressif/frameworks/esp-idf-v5.5/components/heap F:/Espressif/frameworks/esp-idf-v5.5/components/log F:/Espressi
f/frameworks/esp-idf-v5.5/examples/get-started/hello_world/main F:/Espressif/frameworks/esp-idf-v5.5/components/mbedtls
F:/Espressif/frameworks/esp-idf-v5.5/components/newlib F:/Espressif/frameworks/esp-idf-v5.5/components/partition_table F
:/Espressif/frameworks/esp-idf-v5.5/components/pthread F:/Espressif/frameworks/esp-idf-v5.5/components/soc F:/Espressif/
frameworks/esp-idf-v5.5/components/spi_flash F:/Espressif/frameworks/esp-idf-v5.5/components/xtensa
-- Configuring done (0.9s)
-- Generating done (0.2s)
-- Build files have been written to: F:/Espressif/frameworks/esp-idf-v5.5/examples/get-started/hello_world/build
Running ninja in directory F:\Espressif\frameworks\esp-idf-v5.5\examples\get-started\hello_world\build
Executing "ninja all"...
[4/487] Generating ../../partition_table/partition-table.bin
Partition table binary generated. Contents:
*******************************************************************************
# ESP-IDF Partition Table
# Name, Type, SubType, Offset, Size, Flags
nvs,data,nvs,0x9000,24K,
phy_init,data,phy,0xf000,4K,
factory,app,factory,0x10000,1M,
*******************************************************************************
[61/487] Building C object esp-idf/esp_hw_support/CMakeFiles/__idf_esp_hw_support.dir/esp_clk.c.obj
```

As shown in the figure, this is what a successful compilation looks like:

```
-- Build files have been written to: F:/Espressif/frameworks/esp-idf-v5.5/examples/get-started/hello_world/build/bootlo
der
[483/487] Generating binary image from built executable
esptool.py v4.9.1
Creating esp32 image...
Merged 2 ELF sections
Successfully created esp32 image.
Generated F:/Espressif/frameworks/esp-idf-v5.5/examples/get-started/hello_world/build/hello_world.bin
[484/487] C:\Windows\system32\cmd.exe /C "cd /D F:\Express....5/examples/get-started/hello_world/build/hello_world.bin"
hello_world.bin binary size 0x284c0 bytes. Smallest app partition is 0x100000 bytes. 0xda840 bytes (85%) free.
[117/118] Generating binary image from built executable
esptool.py v4.9.1
Creating esp32 image...
Merged 2 ELF sections
Successfully created esp32 image.
Generated F:/Espressif/frameworks/esp-idf-v5.5/examples/get-started/hello_world/build/bootloader/bootloader.bin
[118/118] C:\Windows\system32\cmd.exe /C "cd /D F:\Express....s/get-started/hello_world/build/bootloader/bootloader.bin"
Bootloader binary size 0x6680 bytes. 0x980 bytes (8%) free.
[487/487] Completed 'bootloader'
Project build complete. To flash, run:
 idf.py flash
or
 idf.py -p PORT flash
or
 python -m esptool --chip esp32 -b 460800 --before default_reset --after hard_reset write_flash --flash_mode dio --flash
_size 2MB --flash_freq 40m 0x1000 build\bootloader\bootloader.bin 0x8000 build\partition_table\partition-table.bin 0x100
00 build\hello_world.bin
or from the "F:\Espressif\frameworks\esp-idf-v5.5\examples\get-started\hello_world\build" directory
 python -m esptool --chip esp32 -b 460800 --before default_reset --after hard_reset write_flash "@flash_args"
PS F:\Espressif\frameworks\esp-idf-v5.5\examples\get-started\hello_world>
```

Can be run after compilation is complete

`idf.py build flash monitor`

Download and view the log information of serial port output

If you find the download speed slow, you can run the following command and add the -b parameter, followed by the download baud rate.

`idf.py -b 2000000 build flash monitor`

If you don't want to download all partitions again, run the following command, and only the app partition will be downloaded

`idf.py build app-flash monitor`

If you want to merge the generated bins into a single bin file, run

`idf.py merge-bin` will generate a merged-binary.bin file in the build folder, and using this file allows other devices to download the complete firmware in one go

When using different models of chips, the following are examples of commands for setting up the chips:

`idf.py set-target esp32` Set the target chip to esp32

`idf.py set-target esp32s2` Set the target chip to esp32s2

`idf.py set-target esp32s3` Set the target chip to esp32s3

`idf.py set-target esp32c3` Set the target chip to esp32c3

`idf.py set-target esp32p4` Set the target chip to esp32p4

If multiple ESP32 development boards are plugged into the computer, you need to distinguish the ports. You can open the Device Manager to view the communication ports. For example, if the port of my board is COM5, you need to run:

`idf.py -p COM5 build flash monitor`

When moving a project to another path, the build folder needs to be deleted

Do not place the project in a Chinese directory path!

Develop a good habit of deleting the build folder! It can solve various minor issues

4. Compile Xiaozhi

1. Download the source code

Go to Xiaoge's code homepage:https://github.com/78/xiaozhi-esp32

Click Code - Download ZIP

If Git is installed on your computer, it is recommended to use the command `git clone https://github.com/78/xiaozhi-esp32`, as downloading the zip file may sometimes cause issues

After the download is complete extract it to any directory for later use

Enter the terminal, open IDF 5.5, or double-click ESP IDF 5.5 Powershell on the desktop

The directory I extracted is different from yours. Please each enter the directory you just extracted:

```
cd J:\xiaozhi-esp32-main_2\xiaozhi-esp32-main
```

2. Set the chip type

The default compilation chip is ESP32, so if your own board is S3, you must enter the following command to set the chip to S3

```
idf.py set-target esp32s3
```

If you are using the ESP32C3 chip, you need to enter `idf.py set-target esp32c3` to set the chip type to C3 first, and then go to menuconfig to select the corresponding board type

3. Change the type of the compilation board

The default compiled firmware is for the breadboard. If you need to change the board, you need to enter `idf.py menuconfig`

Enter the Xiaozhi Assistant option

Select Board Type to choose your own board type

Select the corresponding model based on your own board; it's not necessary to choose the Kevin SP V3 board. If you have an ESP32C3 board, you should choose the C3 model and must first run idf.py set-target esp32c3

After selecting the board firmware to be compiled, press the 'S' key on the keyboard to save, then press Esc to exit menuconfig

4. Start to compile

Enter again `idf.py build` to compile

After compilation is completed, enter `idf.py build flash monitor` to perform download and display logs

5. How to modify the wake word

Enter `idf.py menuconfig` to enter the configuration interface

Use the up and down arrow keys on the keyboard to select ESP Speech Recognition

After selecting any wake word, press Enter to confirm, then press 'S' on the keyboard to save and exit

6. How to modify custom wake words (not select, but fully customize)

Use the latest source code

Enter `idf.py menuconfig` to enter the configuration interface, and select Xiaozhi Assistant

Change the Wake Word Implementation Type to the third option

Enter the pinyin of any custom wake word you want in Custom Wake Word, separated by spaces. The pinyin I entered is that of Xiaoke Xiaoke.

There is another one

Then you can compile, download, and test the effect. The effect is as shown in the following log:

7、How to modify the websocket API interface

The latest version 1.6.2 has removed the source code for configuring the websocket interface from menuconfig. The latest source code receives the websocket API interface from the OTA interface and cannot be set manually. You need to switch to version 1.6.0 or lower to do so.

Execute `git checkout v1.6.0`

Enter `idf.py menuconfig` to enter the configuration interface

Use the up and down arrow keys on the keyboard to select Xiaozhi Assistant

Select Websocket and confirm

Enter your own API address, save, exit, and then recompile

8. How to enable the AEC real-time interruption function

Currently, only the LiChuang Shizhanpai S3, ESP-BOX, and ESP32S3 Korvo development boards support speech interruption

In idf.py menuconfig, enter the Xiaozhi Assistant option and check the bottom column

9. Install the serial port driver

If some users' computers do not have serial port drivers and are stuck during the download process, they can try installing the following drivers:

Driver under MAC:

CH341SER_MAC.ZIP
4.23MB

Driver under Windows:

CH343SER.EXE
607.97KB

Solutions to the I2C conflict issue:

If you find the following issues after compiling and programming:

Of course, this issue has been fixed in the latest version of the code

进入idf.py menuconifg->Component config->Audio Codec Device Configuration

Just turn off the first line, then recompile and flash it.